Comparing Lexical, Static-Embedding, and Contextual Drift Signals in Brazilian Portuguese Political Discourse

Anonymous STIL Submission

1

Abstract. *We compare three families of diachronic drift signals in Brazilian Portuguese political discourse: a lexical-profile baseline based on TF-IDF, a static embedding pipeline based on slice-specific Word2Vec models with Orthogonal Procrustes alignment, and a contextual analysis based on a Portuguese BERT checkpoint. The study uses the floor subset of BrPoliCorpus, covering 24 yearly slices from 2000 to 2023 with 428,366 speeches and 63.0 million retained tokens after preprocessing. To make the methods directly comparable, we build a shared panel of 55 lemmas containing method-specific drift candidates, stable controls, and theory seeds. The resulting comparison shows strong disagreement between the lighter baselines, with Spearman correlation of* -0.54 *and zero overlap in their top 15 terms, while contextual BERT remains moderately closer to Word2Vec than to TF-IDF. A filtered contextual panel reduces stable-control leakage and highlights terms such as bloqueio, salário, intervenção, and eleição. Rather than treating one detector as externally validated ground truth, we frame the study as an exploratory comparison of complementary drift sensitivities, interpretability, and computational cost in Brazilian Portuguese political discourse.*

1. Introduction

Political discourse is a productive setting for diachronic NLP because shifts in agenda, coalition structure, institutional conflict, and policy framing leave traces in lexical choice and word usage over time. Recent work has shown that distributional models can capture semantic drift at scale, but it has also made clear that different modeling choices emphasize different aspects of change [Hamilton et al. 2016, Kutuzov et al. 2018]. This is especially relevant in political corpora, where a term may change because of a new policy referent, a rhetorical reframing, a shift in collocational profile, or a reorganization of discourse coalitions rather than because of a neatly isolated lexical sense change.

That ambiguity matters for evaluation. Shared-task benchmarks have improved comparability in lexical semantic change detection, but curated external labels remain scarce, particularly outside the few languages and corpora covered by standard tasks [Schlechtweg et al. 2020]. At the same time, distributional pipelines can react to temporal artifacts, alignment effects, and other confounds if their outputs are interpreted too literally [Dubossarsky et al. 2019]. For our STIL paper, this means that the central question is not which detector is "correct" in an absolute sense, but what each detector appears to capture when applied to the same Portuguese political timeline.

This question is timely for Brazilian Portuguese resources. BrPoliCorpus now offers a large political corpus spanning multiple institutional genres, while Roda Viva provides a complementary interview corpus with cleaner speaker turns and different discourse dynamics [Lima-Lopes 2025, de Miranda Jr. et al. 2024]. We focus here on the

floor subset of BrPoliCorpus as the main discovery corpus and compare a TF-IDF lexical-profile baseline, a slice-specific aligned Word2Vec baseline, and a contextual BERT stage scored on a shared comparison panel. The paper asks three concrete questions: how strongly do the methods agree on the same lemmas, which candidates remain method-specific, and whether the higher-cost contextual stage adds enough interpretive value to justify its cost. That comparison is especially useful in Portuguese, where large political corpora and paper-ready comparative drift analyses are still much scarcer than in the better-studied English-centered literature.

2. Related Work

Distributional semantic change detection has evolved from co-occurrence-profile comparisons to aligned embeddings and then to contextual representations [Kutuzov et al. 2018]. Aligned static embeddings remain attractive because they are comparatively efficient and produce interpretable neighborhood movements, but they also compress all usages of a word into a single representation. This can blur cases in which older and newer usages coexist rather than replacing one another cleanly.

Evaluation is the harder problem. The field has benefited from standardized shared tasks and annotation protocols for lexical semantic change [Schlechtweg et al. 2020], yet many real-world studies still operate without external semantic ground truth. In such settings, negative controls, corpus-sensitive diagnostics, and cautious interpretation are essential because apparent change can emerge from temporal referencing, alignment noise, or shifts in genre and discourse structure rather than from semantic drift alone [Dubossarsky et al. 2019]. This concern is central to our paper and directly motivates the use of stable controls and agreement analysis.

Contextual models offer a partial response to the aggregation problem. By encoding token-level usages, BERT-style representations can surface variation that static spaces smooth over, and prior work has shown that contextualized representations can support meaningful lexical semantic change analysis when paired with usage-sensitive comparison procedures [Giulianelli et al. 2020, Devlin et al. 2019]. Our paper does not position contextual BERT as a final arbiter. Instead, we place it beside TF-IDF and aligned Word2Vec on the same Portuguese political panel so that agreement, disagreement, interpretability, and computational tradeoffs can be examined directly. For Brazilian Portuguese, that comparative angle also fills a descriptive gap: the relevant corpora are only now becoming available at a scale that supports this kind of controlled multi-method analysis in political discourse [Lima-Lopes 2025, de Miranda Jr. et al. 2024].

3. Corpus and Experimental Setup

The experiments use the floor subset of BrPoliCorpus as the main diachronic panel. The frozen baseline contains 24 yearly slices from 2000 to 2023, 428,366 prepared speeches, and 63,036,642 retained tokens after normalization, content-token filtering, and lemma repair. The paper-facing comparison operates on the frozen `content_lemma` representation, preserved as 24 yearly text files totaling 538.5 million characters. Table 1 summarizes the corpus totals and the quartile structure of yearly variation.

At the slice level, yearly document counts range from 3,041 speeches in 2000 to 26,868 in 2015, with a median of 16,751.5 documents per year. Retained token volume

Table 1. Corpus summary for the frozen BrPoliCorpus floor discovery panel, which contains Brazilian Chamber floor speeches. Yearly slice statistics are reported as p25/median/p75 across the 24 yearly files; token and character counts are shown in millions.

Dataset	Period	Slices	Corpus totals	Yearly slices (Q1/Med/Q3)
			428,366 speeches	14,648 / 16,752 / 21,987 speeches
			63.0M tokens	2.00 / 2.62 / 3.45M tokens
			538.5M chars	16.9 / 22.3 / 29.7M chars
BrPoliCorpus floor	2000–2023	24	225,484 lemmas	39,456 / 50,269 / 58,637 lemmas

ranges from 630,426 tokens in 2000 to 4,171,779 in 2003, with a median of 2,620,480 tokens. The corresponding yearly text files range from 5.4 million characters in 2000 to 36.0 million in 2003, with a median of 22.3 million characters. The lemma statistics table preserved for the frozen run contains 1,184,808 slice–lemma rows and 225,484 unique lemmas overall; yearly vocabulary size ranges from 28,768 lemmas in 2000 to 74,578 in 2011, with a median of 50,269. Document volume is high throughout the panel, but token volume and vocabulary size vary enough to motivate explicit controls for frequency, slice presence, and candidate eligibility.

We deliberately do not merge BrPoliCorpus and Roda Viva into a single time-aligned corpus. The present paper uses BrPoliCorpus floor as the sole discovery panel, while Roda Viva remains a complementary corpus for later triangulation under separate preprocessing and temporal assumptions.

4. Methodology

The comparison begins with a frozen Word2Vec baseline built from yearly slices, skip-gram training, and Orthogonal Procrustes alignment across time, following the standard family of aligned diachronic embedding approaches [Hamilton et al. 2016]. We then add a lexical baseline based on TF-IDF change between adjacent slices and a contextual stage that embeds sampled in-slice occurrences with the Portuguese checkpoint `rufimelo/bert-large-portuguese-cased-sts`. The contextual analysis extracts layers -1 and -4; the agreement layer keeps -1 as the preferred paper-facing ranking and uses the second layer as a robustness view.

Figure 1 summarizes the paper-facing workflow. The frozen discovery corpus first feeds the two cheaper baselines; their method-led candidates are then combined with stable controls and theory seeds to create the shared panel used for contextual inspection and the final agreement layer.

Rather than compare unrestricted vocabularies, we construct a shared panel of 55 lemmas containing 15 Word2Vec-led drift candidates, 15 TF-IDF-led drift candidates, 20 stable controls, and 5 theory seeds. This design keeps the comparison method-neutral: all three families observe the same candidate set, and disagreement can be interpreted as a property of the signal rather than as an artifact of divergent vocabularies. The post-run agreement layer then derives pairwise rank correlations, top-k overlaps, stable-control leakage diagnostics, and a filtered contextual ranking. Figure 2 visualizes the pairwise agreement structure, while Figure 3 summarizes overlap and rank-distribution statistics.

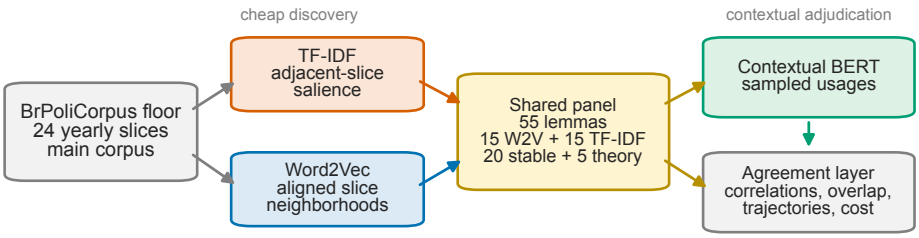

Figure 1. Paper-facing comparative workflow. Cheap baselines nominate candidates; the shared panel combines those candidates with controls and theory seeds before contextual inspection and agreement analysis.

5. Results

The first result is that the two lighter baselines disagree strongly even after they are constrained to the same candidate universe. On the shared panel, Word2Vec and TF-IDF reach Spearman correlation of -0.54 and share zero terms in their top 15 rankings. This inverse association is statistically clear ($p = 2.1 \times 10^{-5}$), which makes the disagreement hard to dismiss as a small-sample accident. The preferred contextual layer partly mediates this split without collapsing onto either baseline. Its correlation with Word2Vec is 0.21 ($p = 0.128$), while its correlation with TF-IDF is 0.12 ($p = 0.365$). These positive associations are directionally informative but not strong enough to justify treating contextual BERT as a substitute for one of the cheaper methods. Figure 2 shows the resulting geometry: the contextual top ranks borrow support from both camps, while the two cheap methods remain sharply separated.

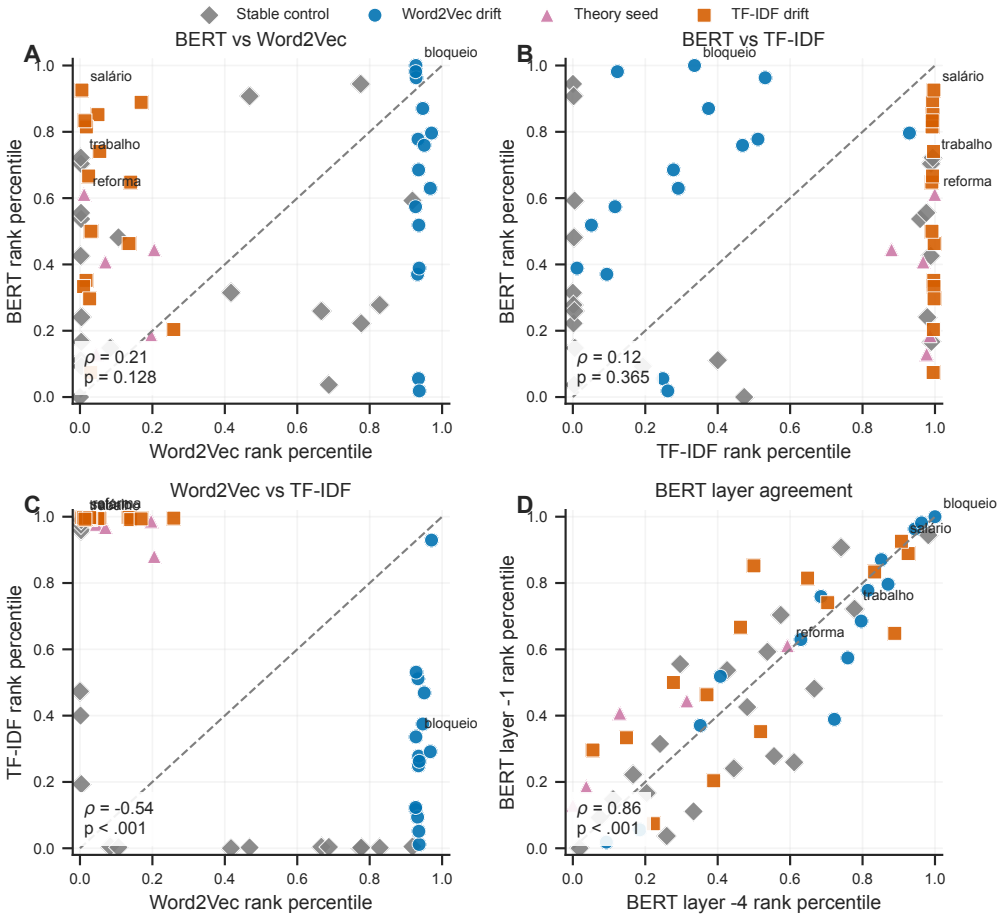

Figure 2. Agreement on the shared panel across TF-IDF, Word2Vec, and the preferred BERT layer; panel (D) compares BERT layers -4 and -1. Annotations report Spearman ρ and two-sided p-values.

The second result is that overlap statistics and bucket-level rank distributions are more informative than raw top-k lists alone. In the preferred BERT layer, the top-15 overlap is 7 terms with Word2Vec and 6 terms with TF-IDF, whereas the baseline-to-baseline overlap remains zero. Figure 3 also shows that selected drift terms are ranked significantly above stable controls by Word2Vec and TF-IDF ($p < .001$ in both cases), while contextual BERT still separates the groups but with weaker margins ($p = .008$). This pattern matches the filtered contextual panel: BERT provides useful adjudication, but the raw contextual ranking still leaks some stable controls and should not be treated as an unrestricted final list. The two contextual extraction layers nevertheless agree strongly with each other (Spearman $= 0.858$), so the broad comparative picture does not depend on a single extraction depth.

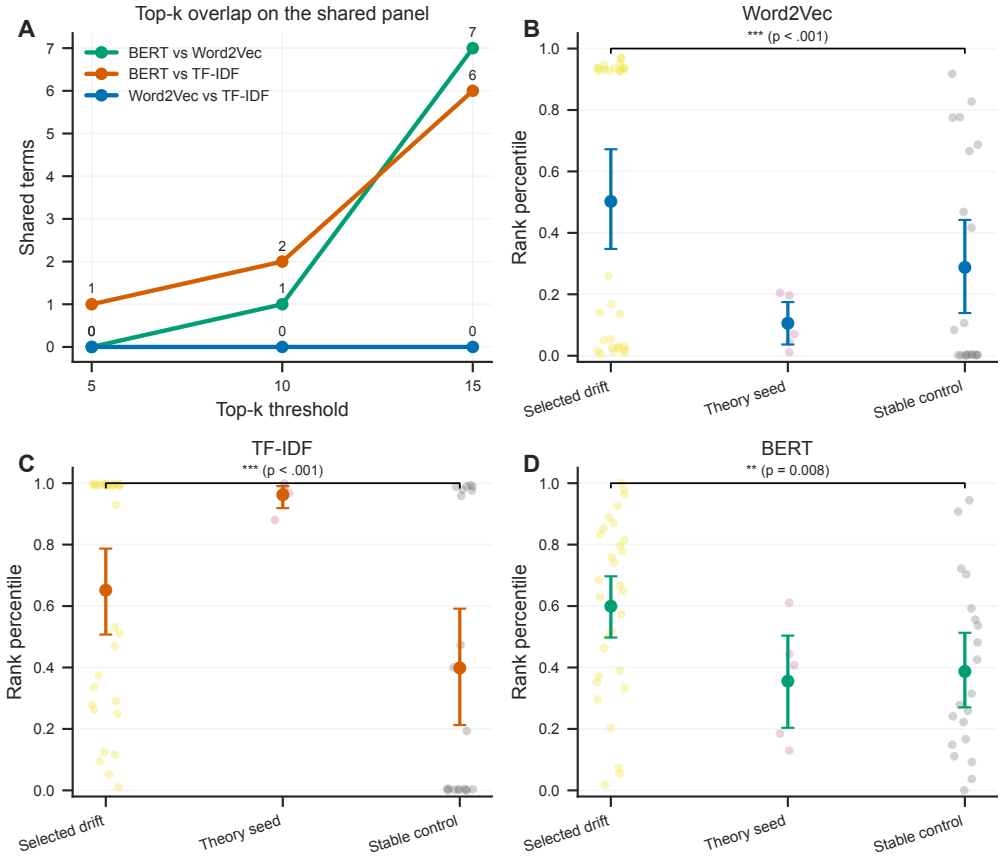

Figure 3. Overlap and rank-distribution summaries on the shared comparison panel. Panel (A) traces top-k overlap; panels (B) through (D) compare rank-percentile distributions for selected drift terms, theory seeds, and stable controls.

6. Discussion

The third result is qualitative rather than purely statistical. Representative term trajectories show that the methods often disagree about when a term moves, not only whether it moves at all. Figure 4 illustrates four cases: a Word2Vec-led term with strong contextual support (*bloqueio*), a TF-IDF-led term with strong contextual support (*salário*), a theory seed whose contextual rank rises despite modest static drift (*reforma*), and a stable-control diagnostic term (*trabalho*) that reveals where contextual leakage remains. These cases show why a comparative analysis is more informative than a single-method ranking.

The disagreement itself is substantively informative. TF-IDF appears most sensitive to reweightings in lexical profile: when a term becomes newly central to yearly agendas, policy disputes, or formulaic framing, its salience can move even if its local semantic neighborhood remains comparatively stable. Word2Vec, by contrast, is better positioned to highlight persistent neighborhood reorganization across slices. Contextual BERT sits between these views. Because it observes token-level usages, it can support either a TF-IDF-led or a Word2Vec-led candidate, but the stable-control leakage table shows that greater context sensitivity does not automatically produce a cleaner unrestricted ranking.

This interpretation fits the term-level evidence. *Salário* behaves like a lexical-

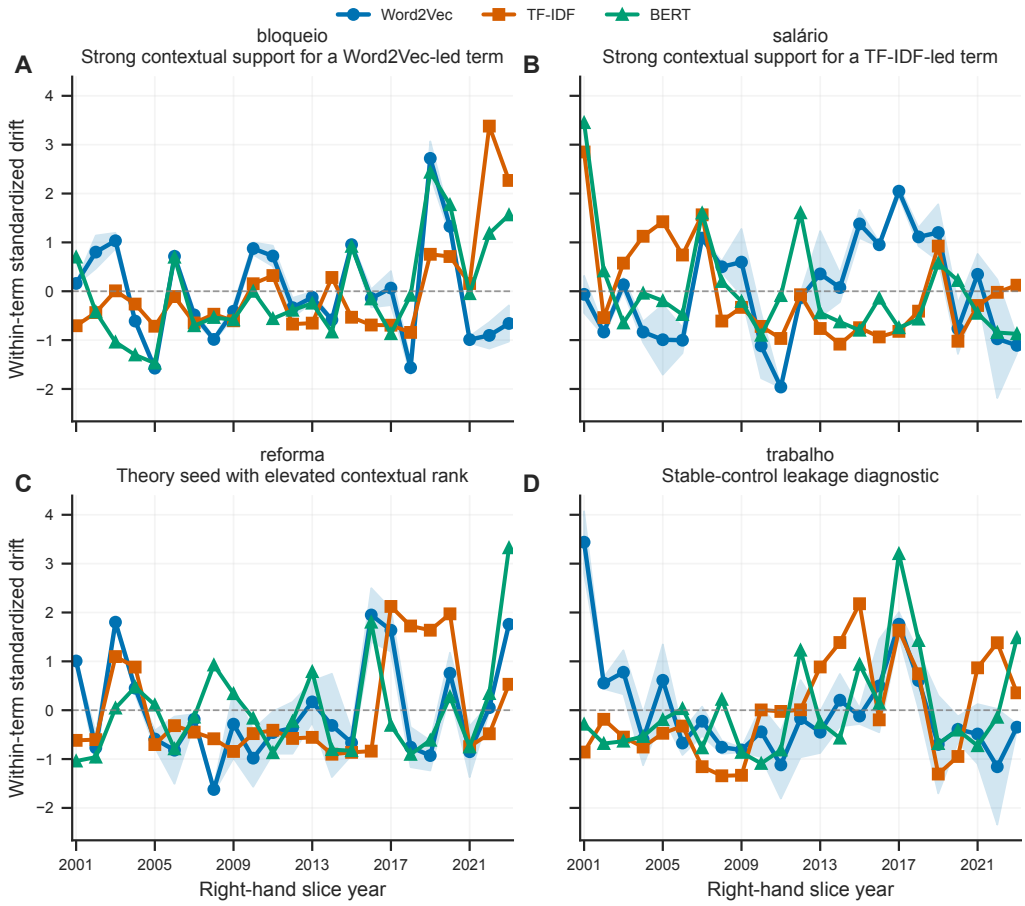

Figure 4. Representative term trajectories across methods. Scores are standardized within term and method to emphasize temporal shape rather than absolute scale.

Table 2. Method scope and runtime on the frozen comparative baseline. The BERT runtime is an approximate wall-clock inferred from run timestamps because no dedicated duration manifest was preserved for that stage.

Method	Signal focus	Scope	Runtime
TF-IDF	Adjacent-slice salience shift	3,221 lemmas	24.7 s
Word2Vec	Aligned neighborhood displacement	3,221 lemmas	22.9 min
Contextual BERT	Usage divergence from sampled contexts	55 lemmas / 82,461 occurrences	$\approx$ 4.12 h CPU

profile case with strong contextual support, suggesting that changes in agenda prominence and usage framing are both at work. *Bloqueio* looks more like a neighborhood-displacement case, where the contextual layer aligns with the embedding-led signal. *Reforma* is useful for a different reason: it is not a top filtered contextual term, but as a theory seed it shows that politically central vocabulary can display contextual differentiation even when it is not the strongest candidate in the cheap rankings. That is precisely why the shared panel includes theory seeds and stable controls alongside method-led drift terms.

The computational comparison in Table 2 further sharpens the story. TF-IDF is nearly instantaneous on the frozen run, Word2Vec is still practical at corpus scale, and contextual BERT is markedly more expensive even after being restricted to the shared panel. Given the modest rank correlations, the current evidence does not support replacing the cheaper methods with BERT. Instead, it supports using contextual modeling selectively as an adjudication and interpretation layer when disagreement is substantively interesting.

7. Conclusion

The frozen baseline supports an exploratory comparative study rather than an argument for a single best detector of semantic change. TF-IDF and Word2Vec highlight clearly different regions of the candidate space, and contextual BERT adds a useful but higher-cost inspection layer that remains distinct from both. For this Portuguese political setting, disagreement is therefore part of the result: it suggests that the three families are sensitive to different mixtures of lexical salience, neighborhood displacement, and usage-level variation.

8. Limitations

The first limitation is evaluative. This paper does not claim external semantic ground truth for BrPoliCorpus floor, so agreement and disagreement are analyzed as comparative evidence rather than as proof that one detector is correct. The second limitation is methodological. Although the filtered contextual list is useful for paper-facing interpretation, the raw contextual ranking still admits some stable controls near the top, which means that contextual sensitivity does not automatically yield a cleaner unrestricted candidate list. The third limitation is temporal granularity: yearly slicing is appropriate for a first comparative baseline, but it compresses within-year shocks, campaign cycles, and short-lived institutional disputes that may matter for political language.

The most plausible next validation-oriented extension is *PTPARL-V*, but we do not use it here as ground truth because its intervention-level labels still require explicit deduplication and aggregation rules before they can serve as a separate noisy supervision layer. In the same spirit, a symbolic companion analysis, such as selected NILC-Metrix-style features, should be treated as an interpretive layer for separating lexical-semantic drift from rhetorical, stylistic, or discourse-organizational change, not as a final validator by itself.

9. Ethics Statement

The study analyzes public institutional political speech and reports only aggregate corpus statistics, ranked lemmas, and usage-level comparative signals. We do not use the models to profile individual speakers or to infer private attributes. Even so, political corpora encode unequal representation, partisan conflict, and occasionally harmful rhetoric, so drift scores should not be read as normative judgments, direct evidence of speaker intent, or standalone claims about social meaning. The outputs in this paper are exploratory comparative signals whose interpretation depends on corpus composition, preprocessing choices, and the distinction between lexical salience, contextual variation, and broader discourse change.

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